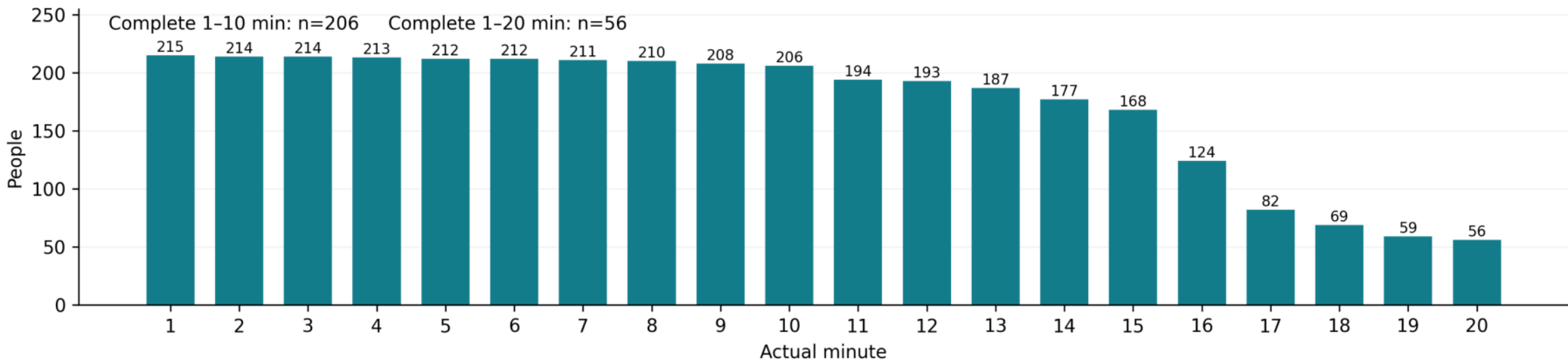


Figure 1 | Observation support and first recorded codes

A Numeric observations at each actual minute



B First recorded code counts

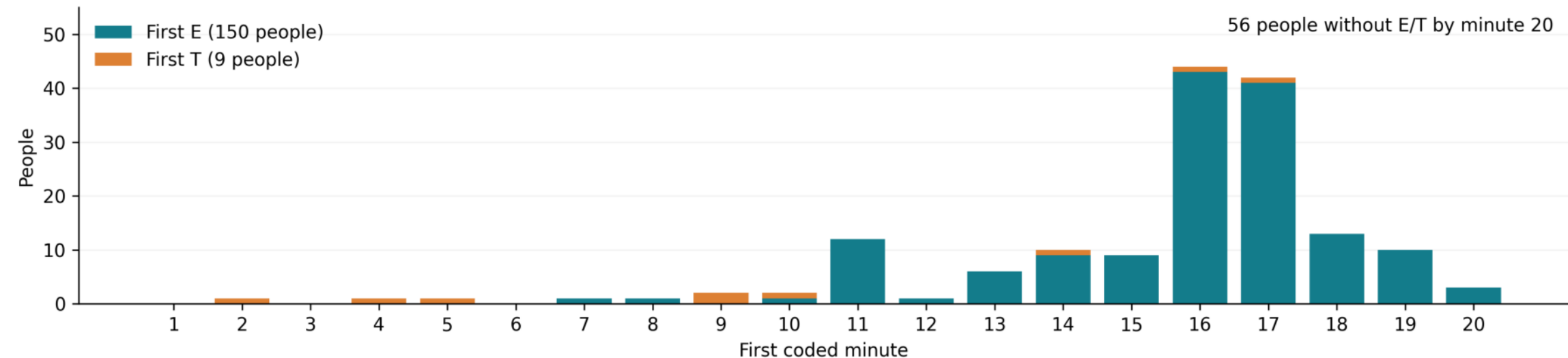
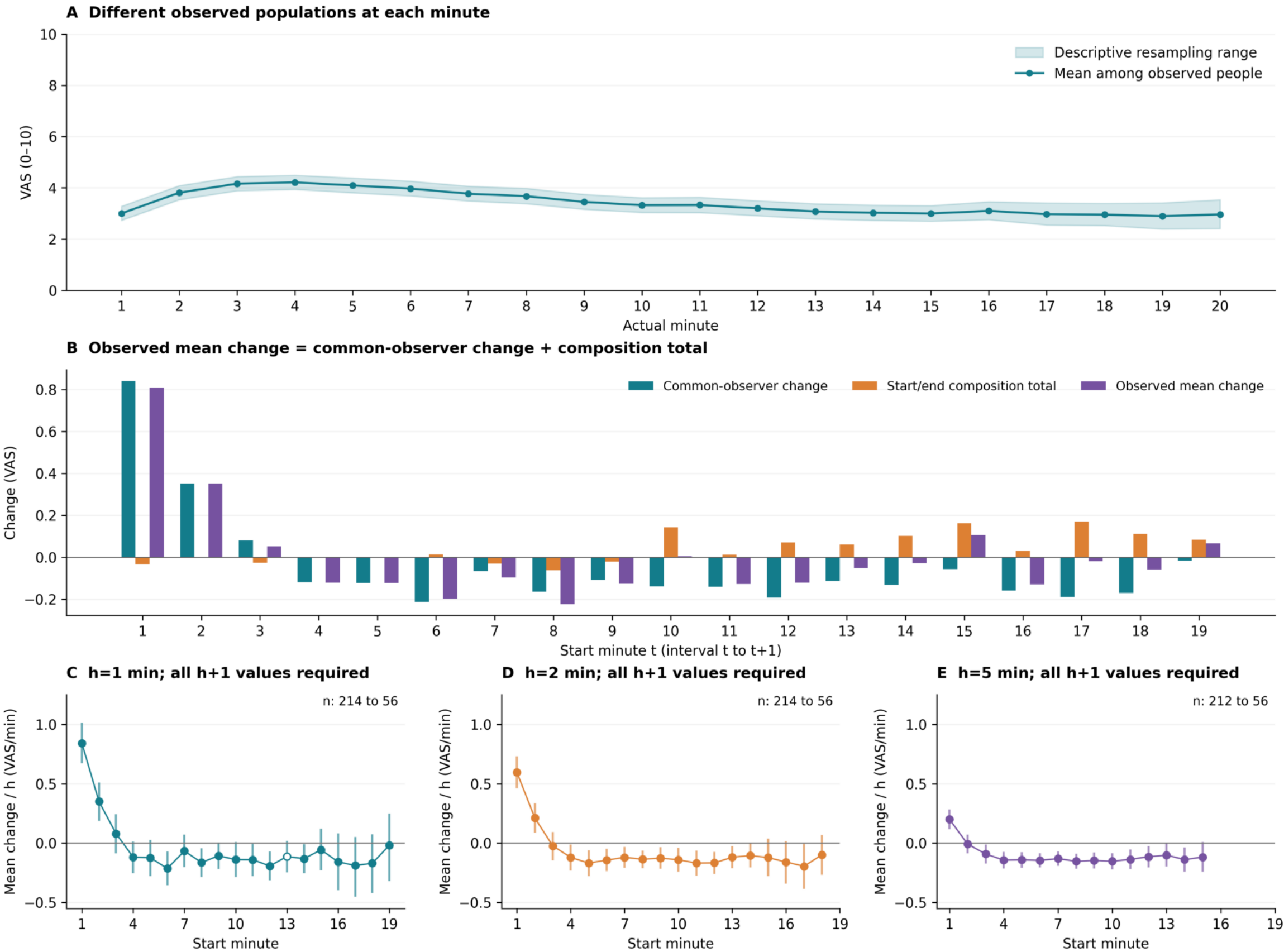


Figure 2 | Observed ratings, composition and finite changes



Ranges: q.025-q.975 from 200,000 whole-person draws; descriptive, not calibrated CIs. Hollow: MC target unmet.

Figure 3 | Recorded symptom and perceived-region context

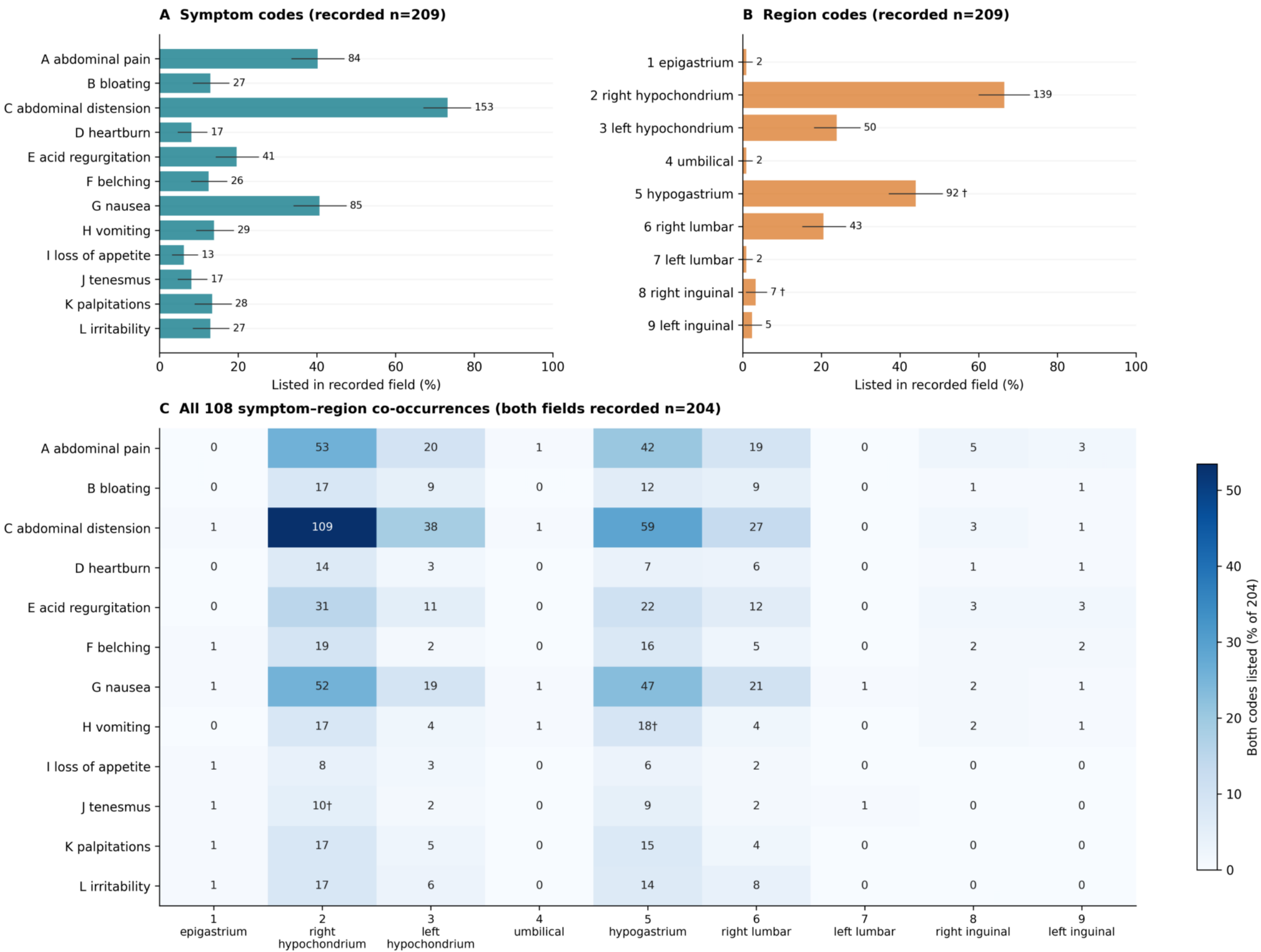
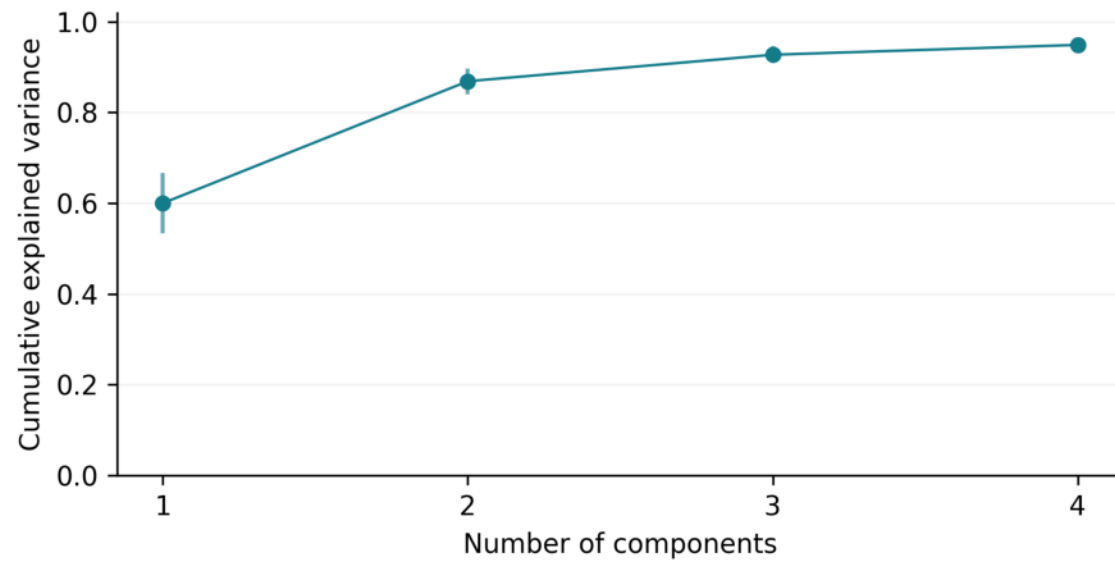
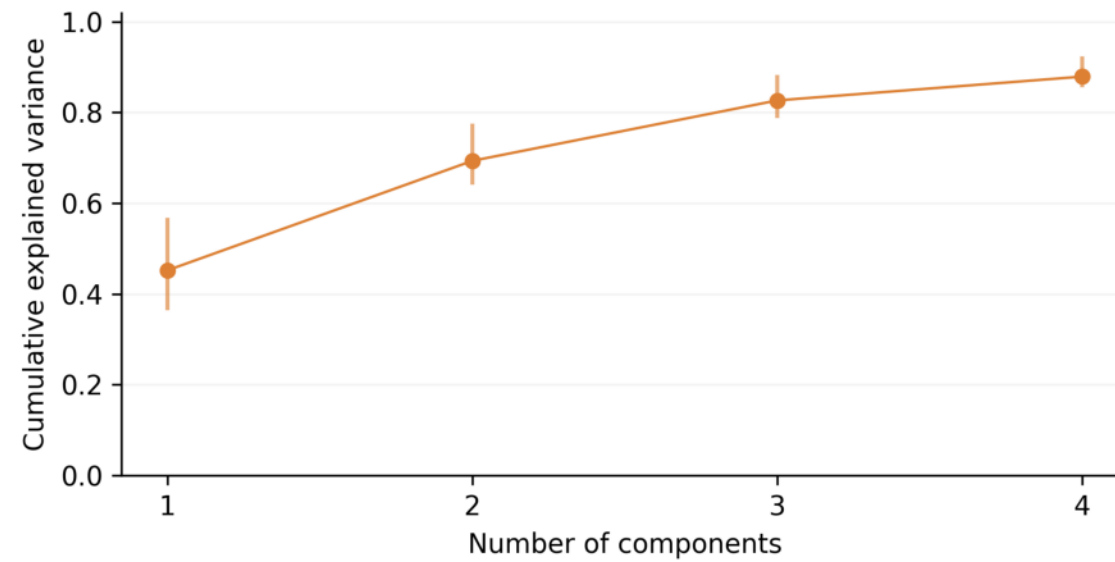


Figure 4 | Complete-curve FPCA

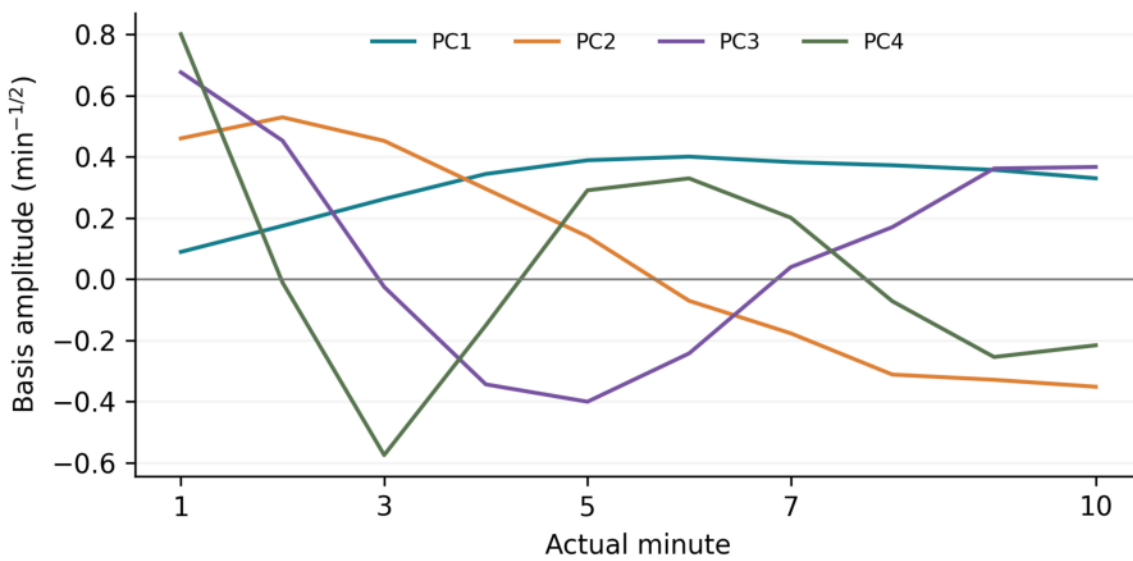
A 1-10 min; n=206



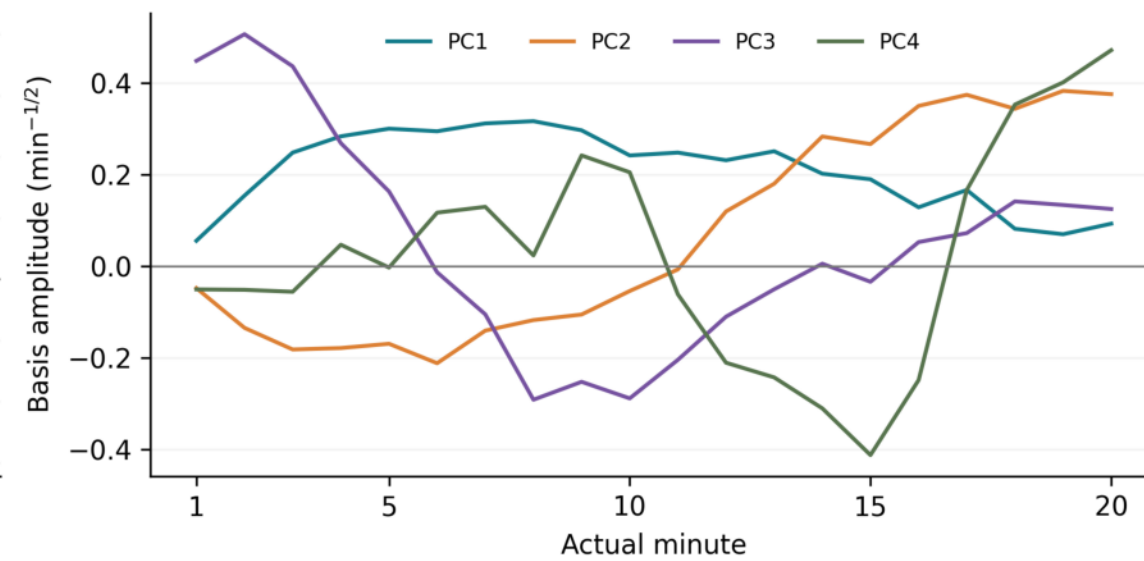
B 1-20 min; n=56



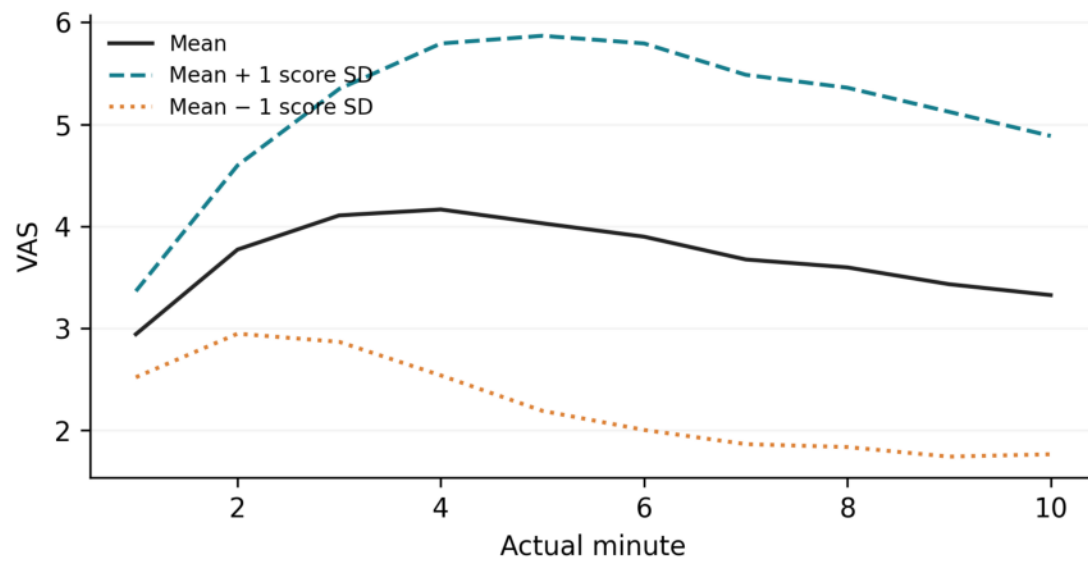
C Weighted bases, 1-10 min



D Weighted bases, 1-20 min



E PC1 model mode (not a CI), n=206



F PC2 model mode (not a CI), n=206

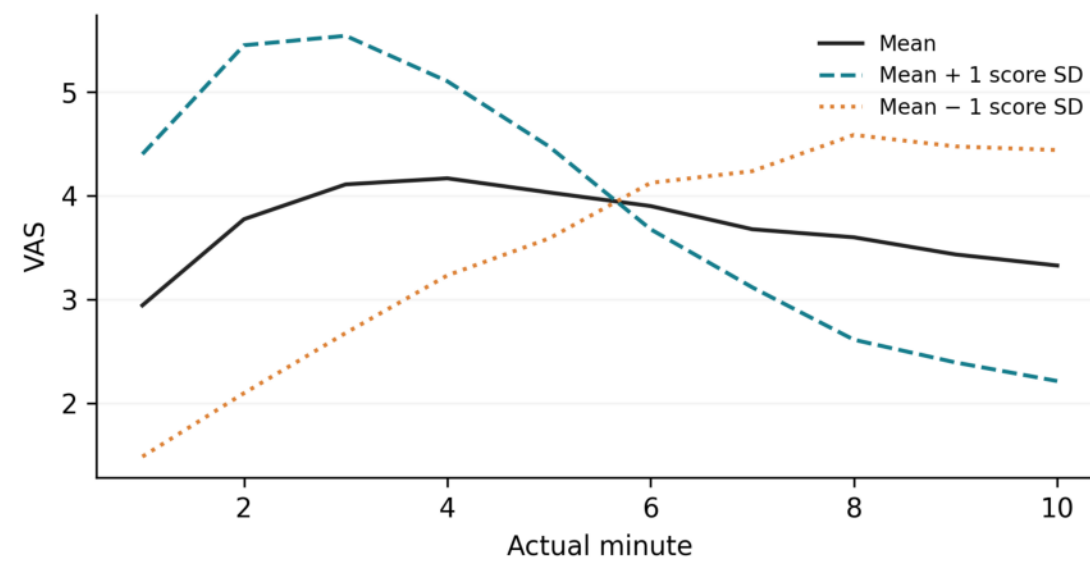
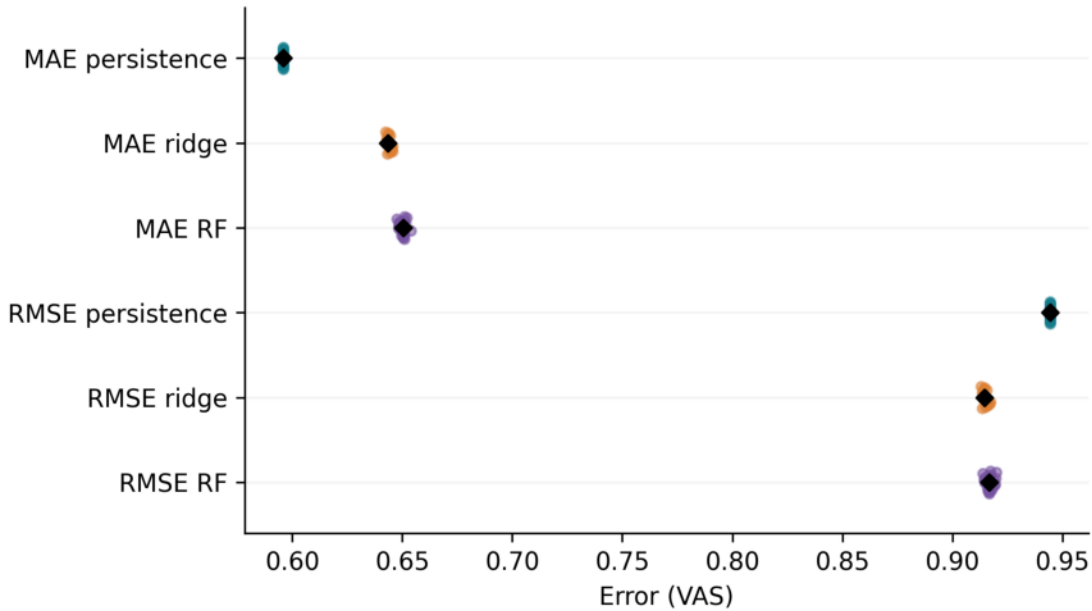
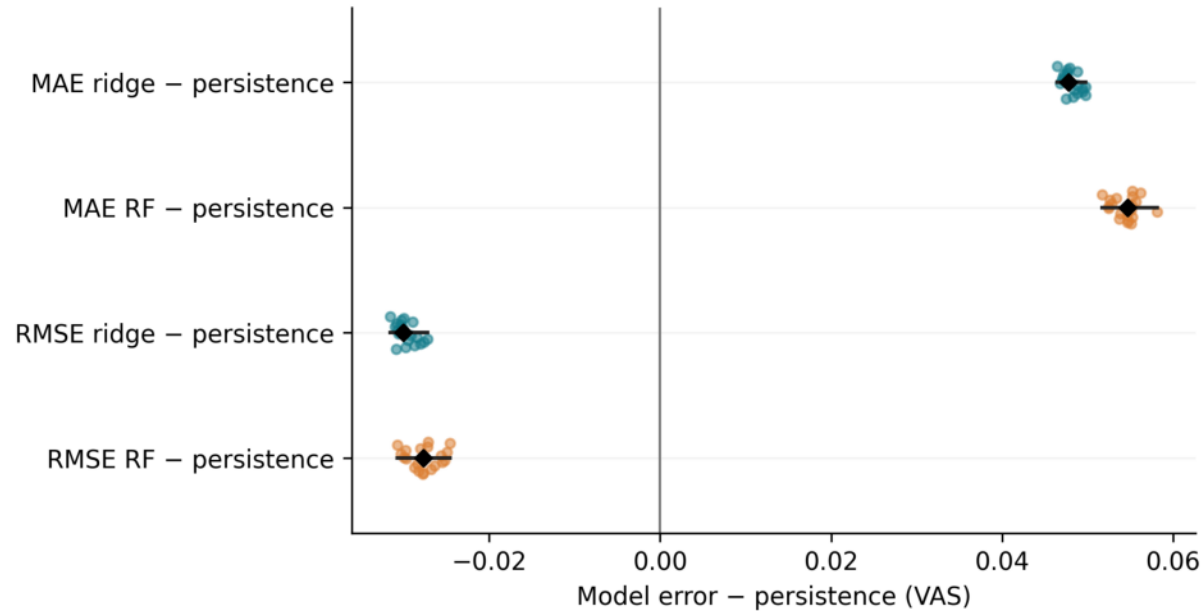


Figure 5 | Internal future-rating evaluation

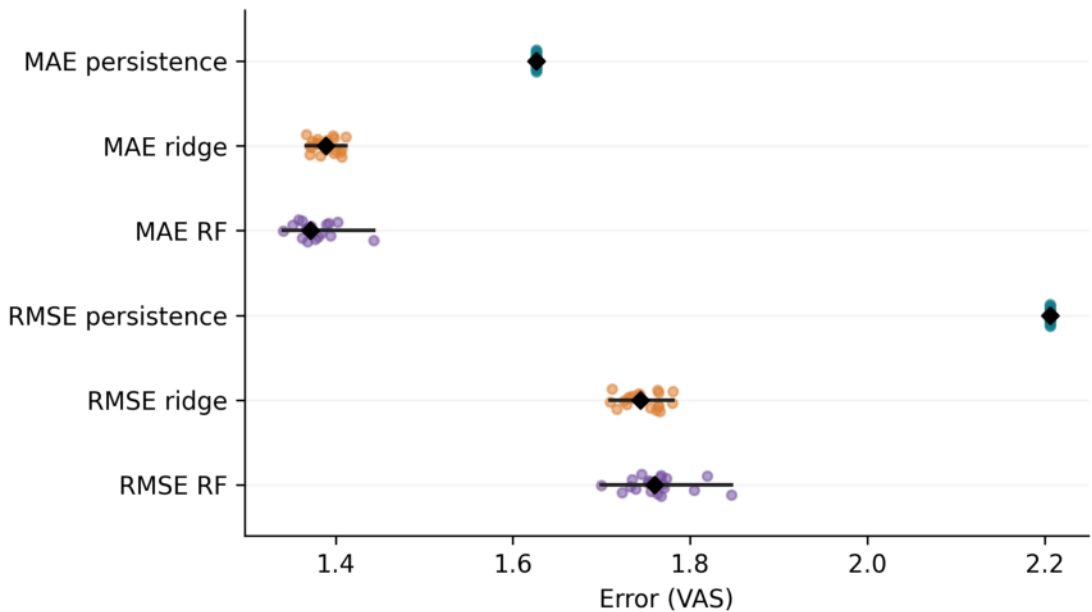
A Next rating: 213 people / 2,781 windows window equal



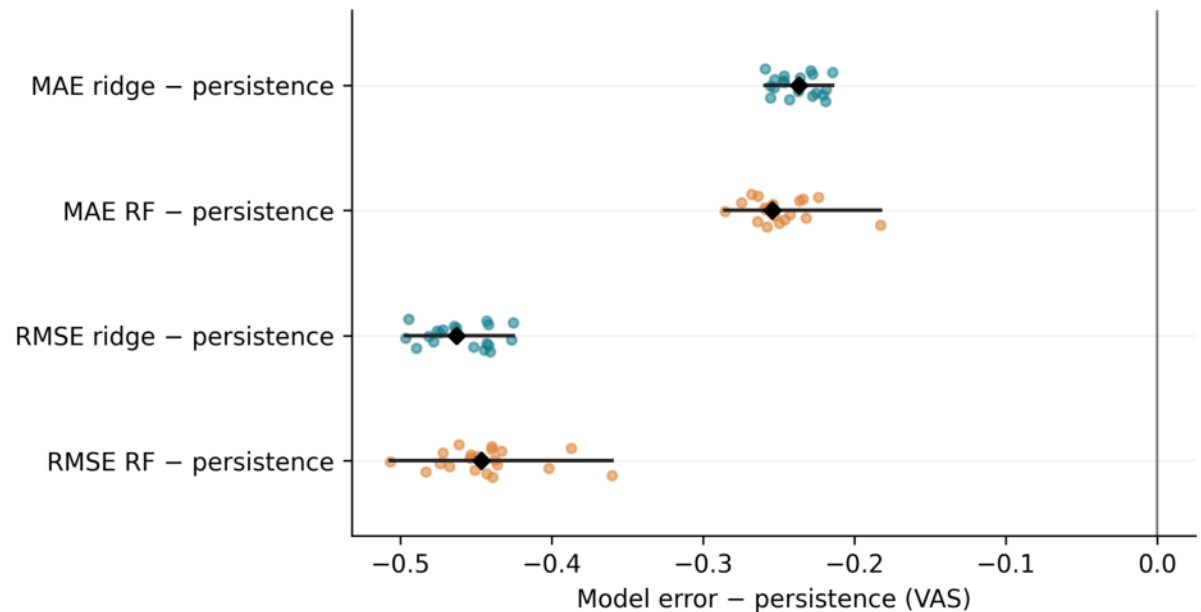
B Same-repeat paired differences



C Minute 5 to 10: 206 people person equal

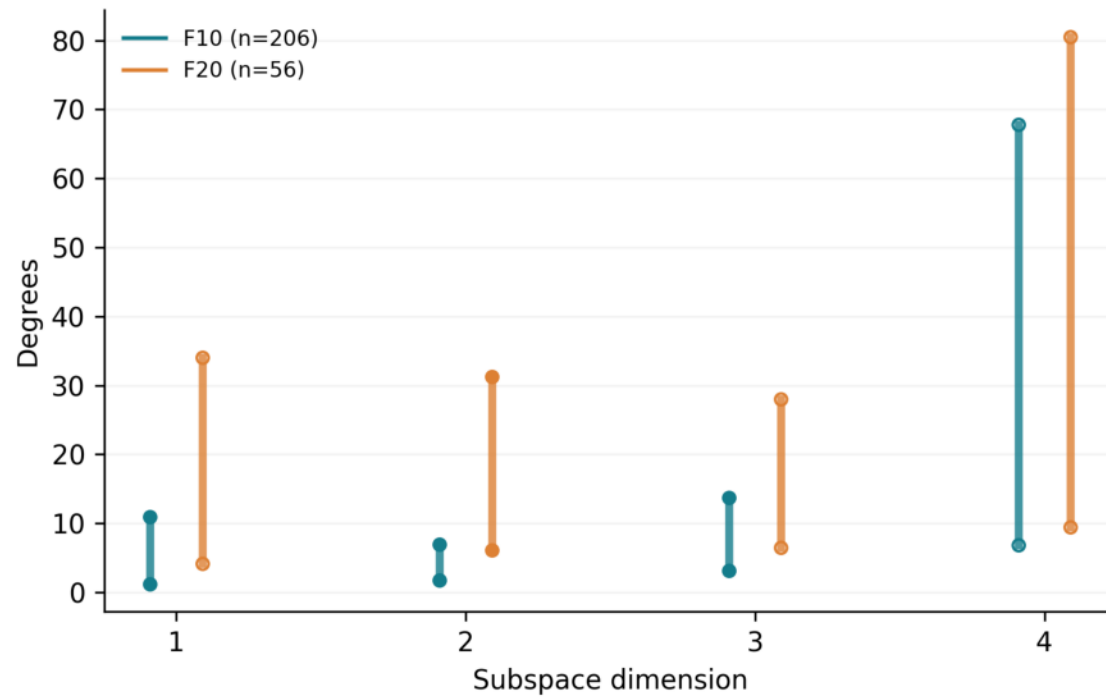


D Same-repeat paired differences

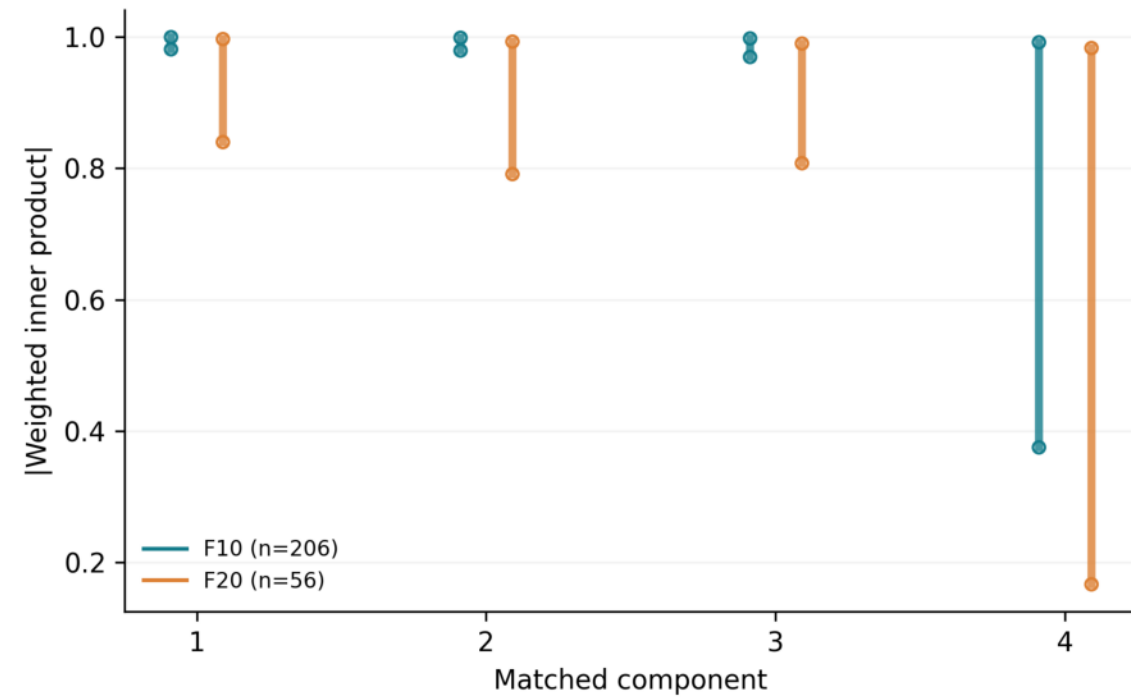


Supplement S1 | FPCA stability and whole-curve reconstruction

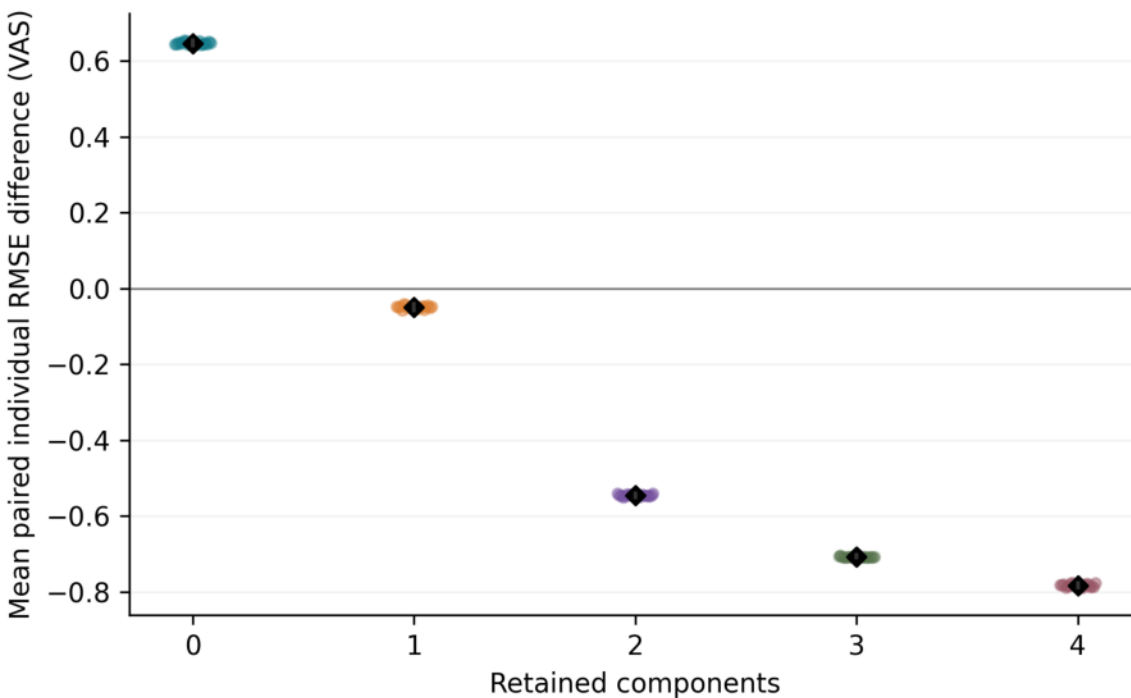
A Maximum subspace angle



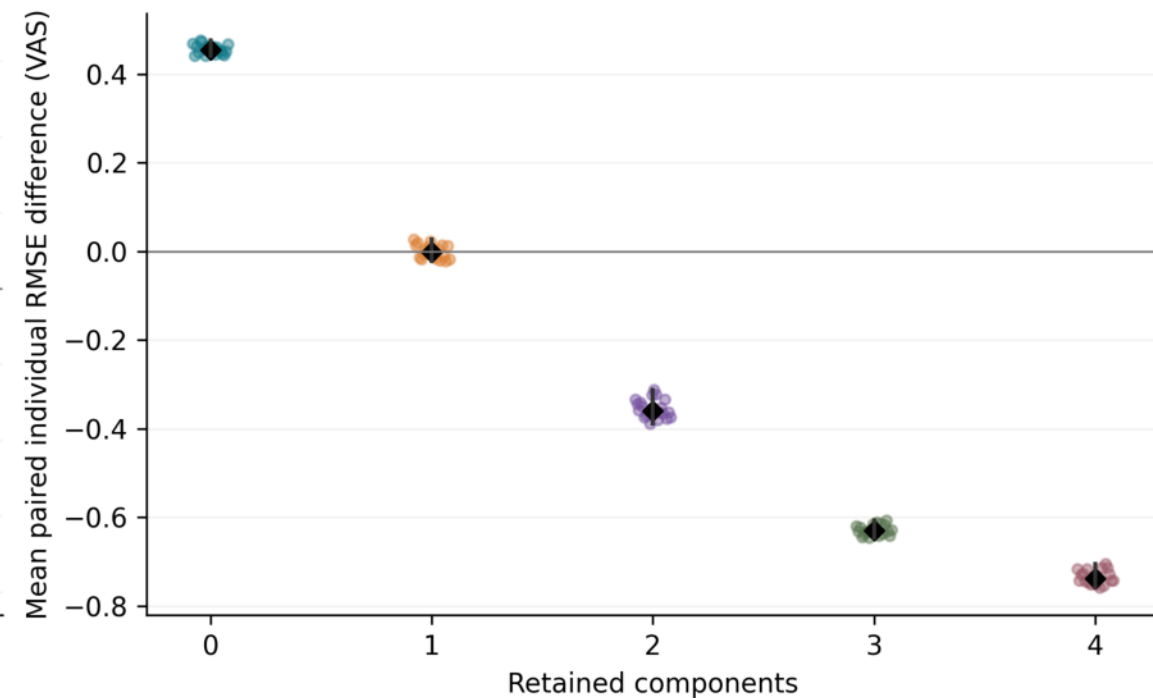
B Matched absolute axis inner product



C F10; all 20 repeats

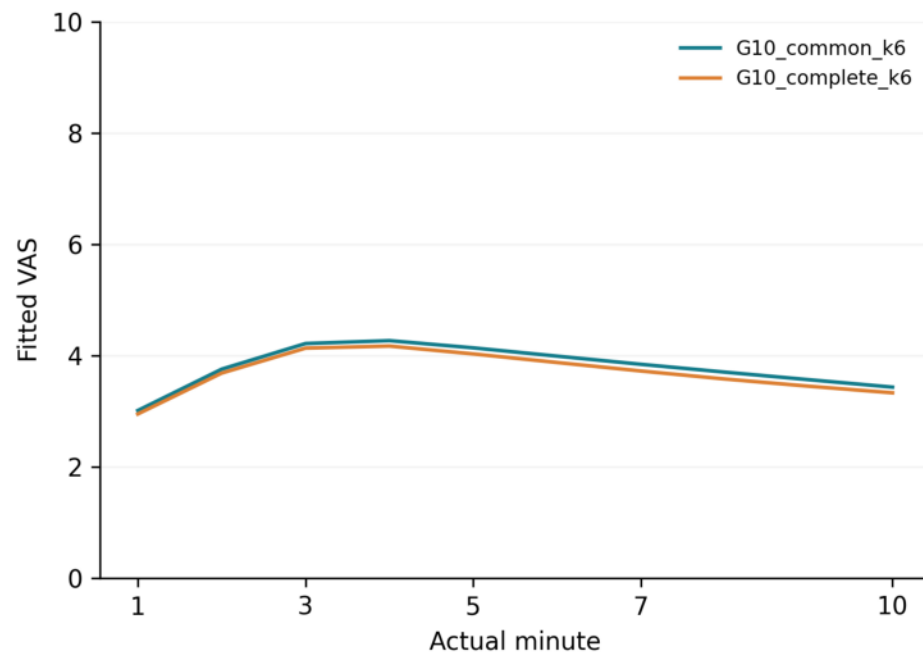


D F20; all 20 repeats

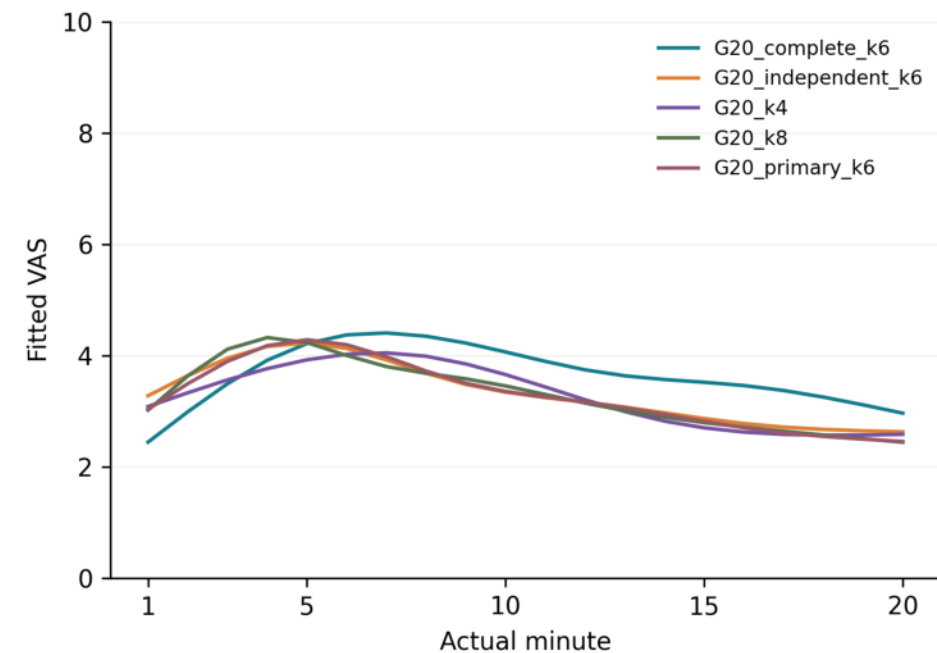


Supplement S2 | GAMM working models and numerical diagnostics

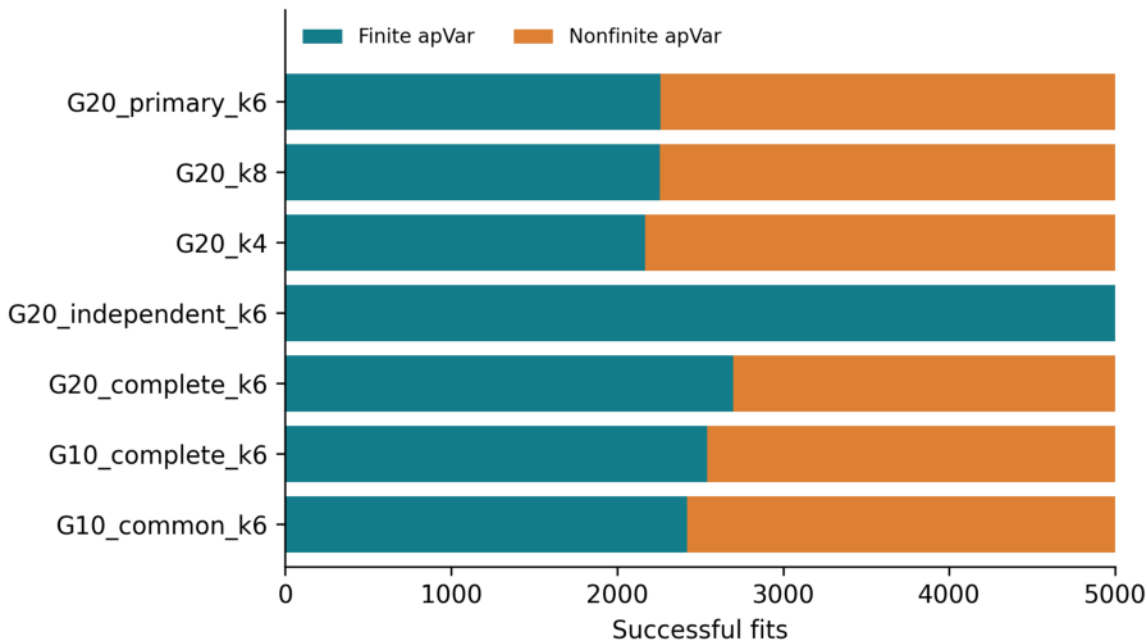
A 10-minute reference curves



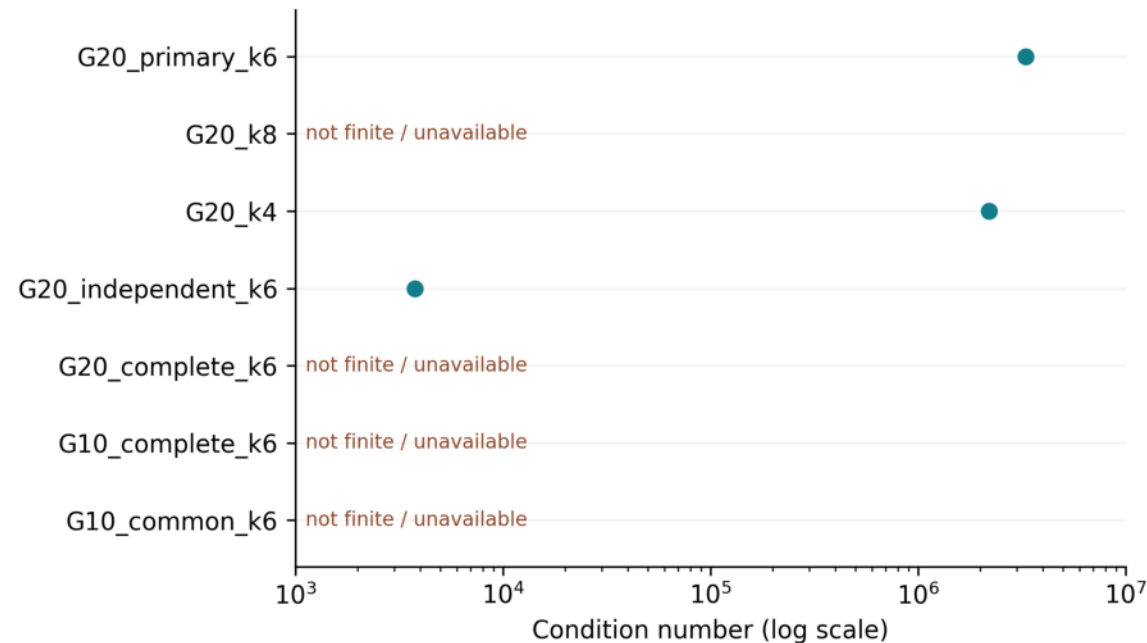
B 20-minute reference curves



C apVar records per 5,000-fit cell

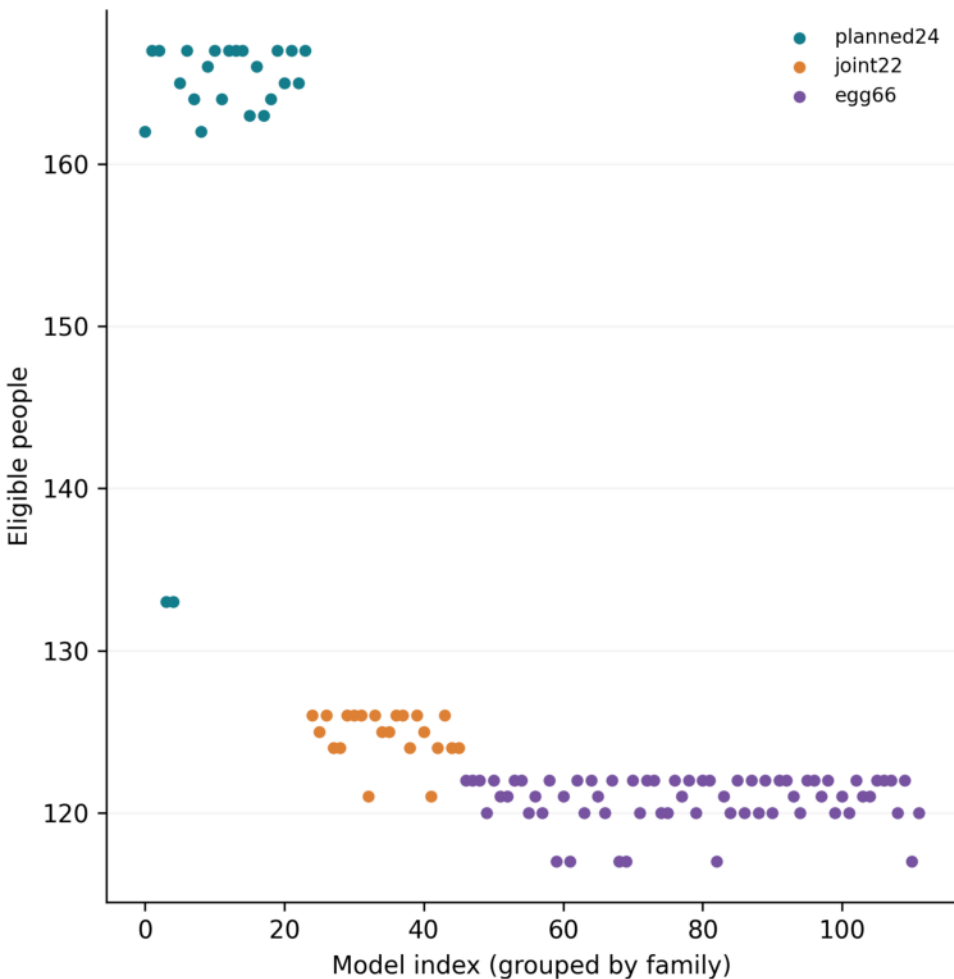


D Approximate covariance conditioning

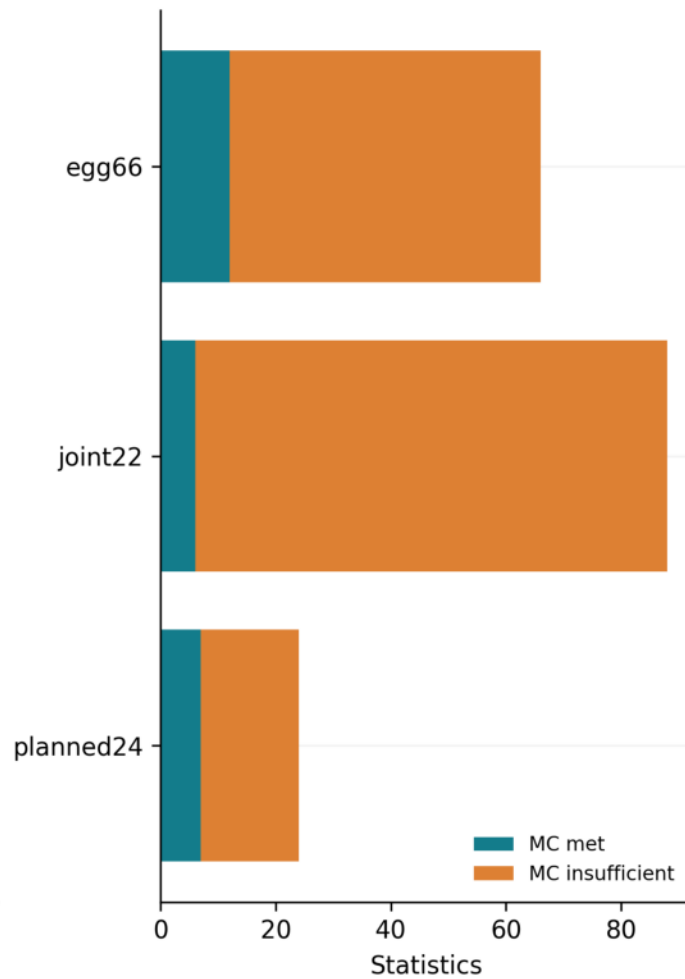


Supplement S3 | Candidate-model support and retained diagnostics

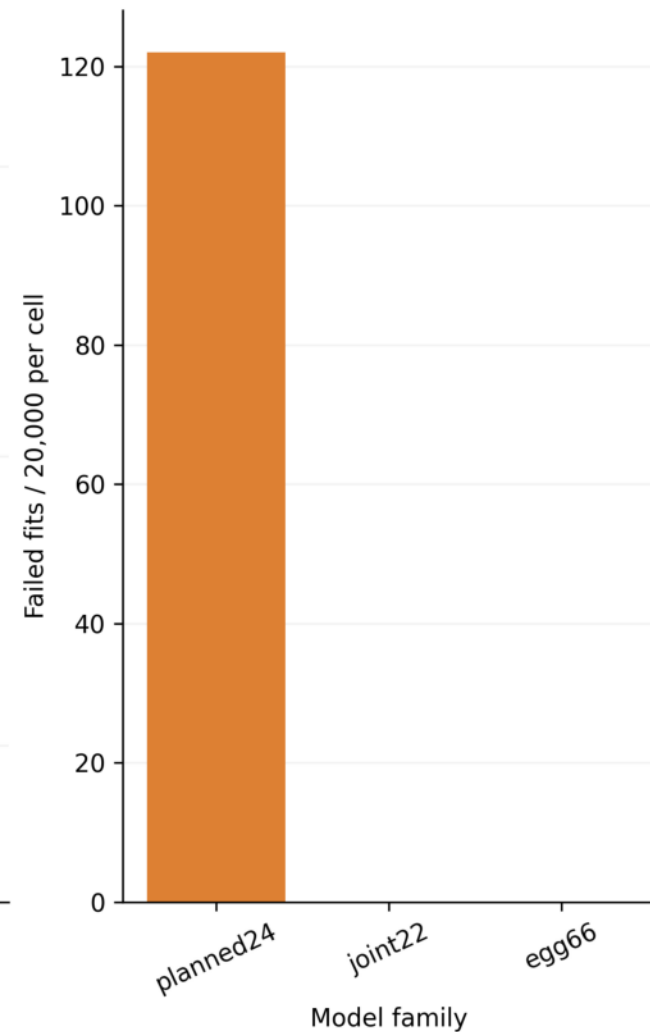
A Subject support in every model cell



B All 178 statistics

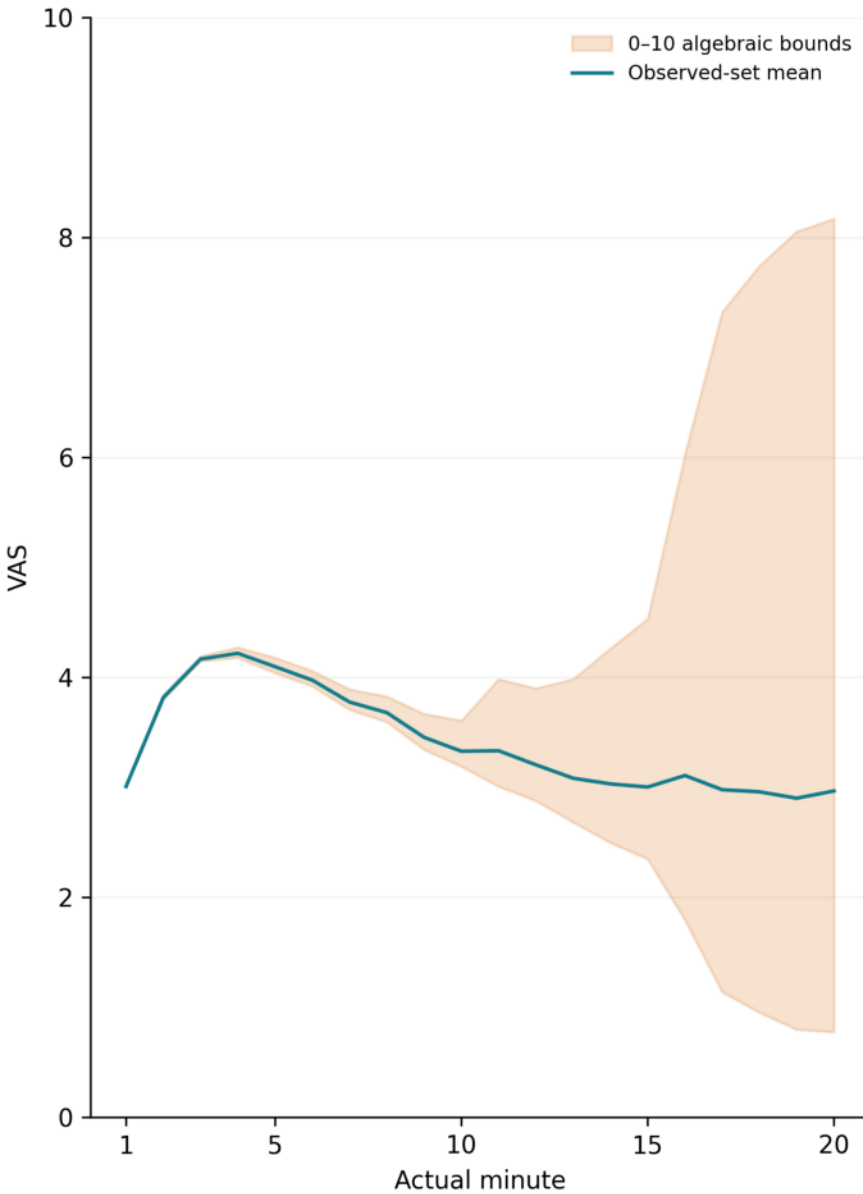


C Rank failures retained

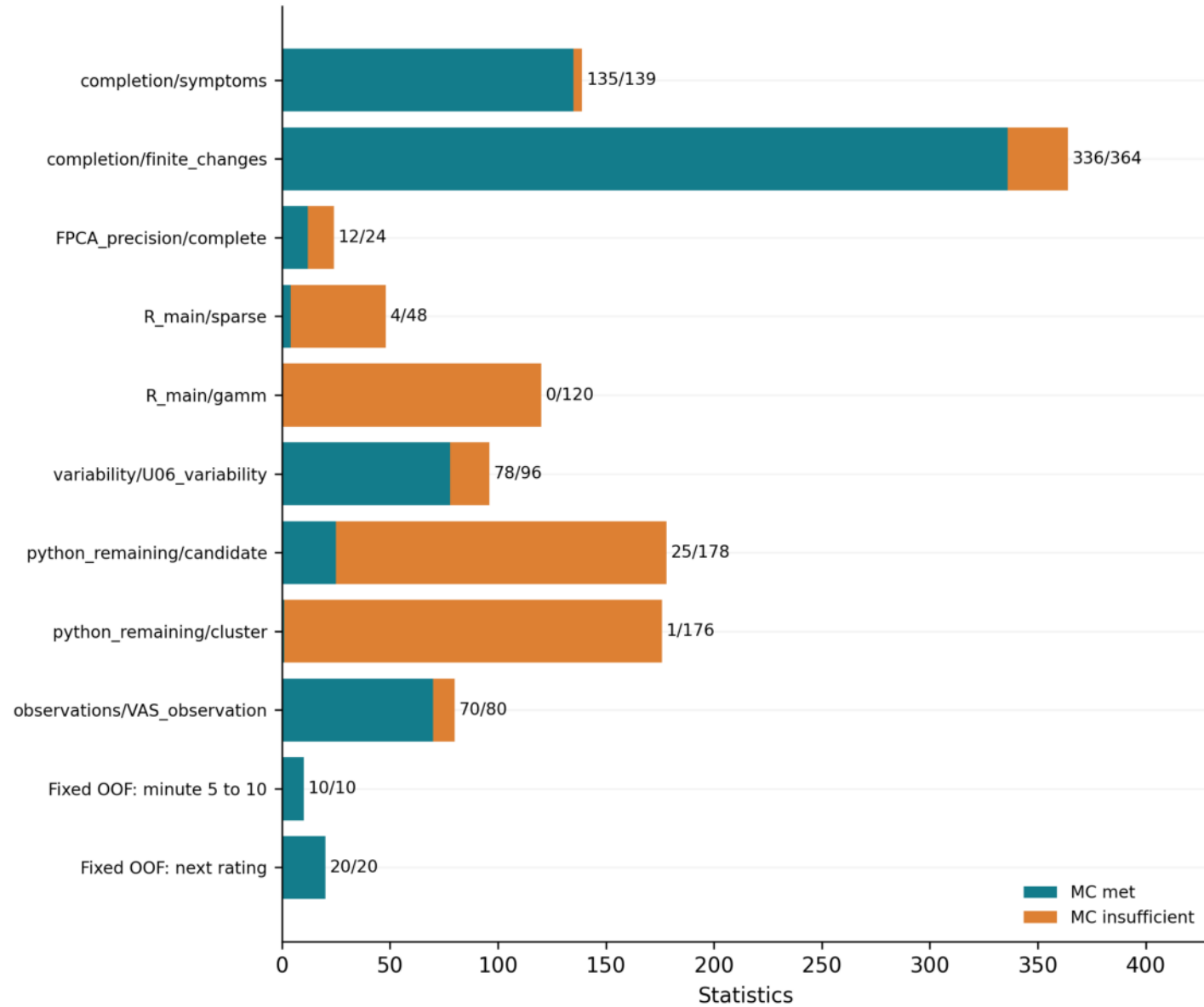


Supplement S4 | Missing-rating algebra and MC status

A Algebraic full-cohort mean bounds



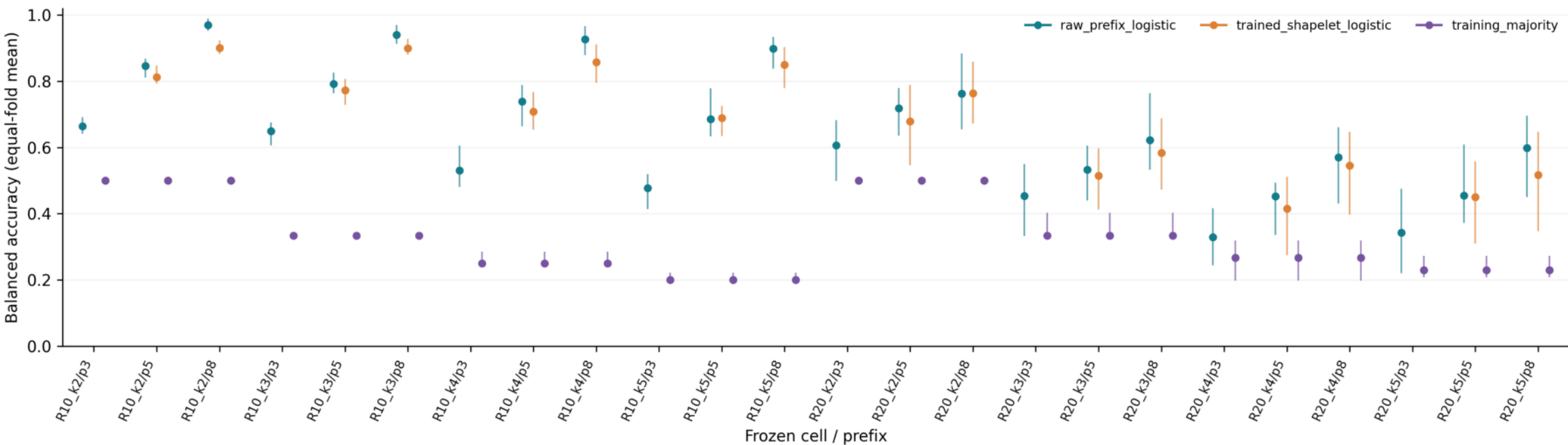
B All 1,255 current statistics; superseded FPCA excluded



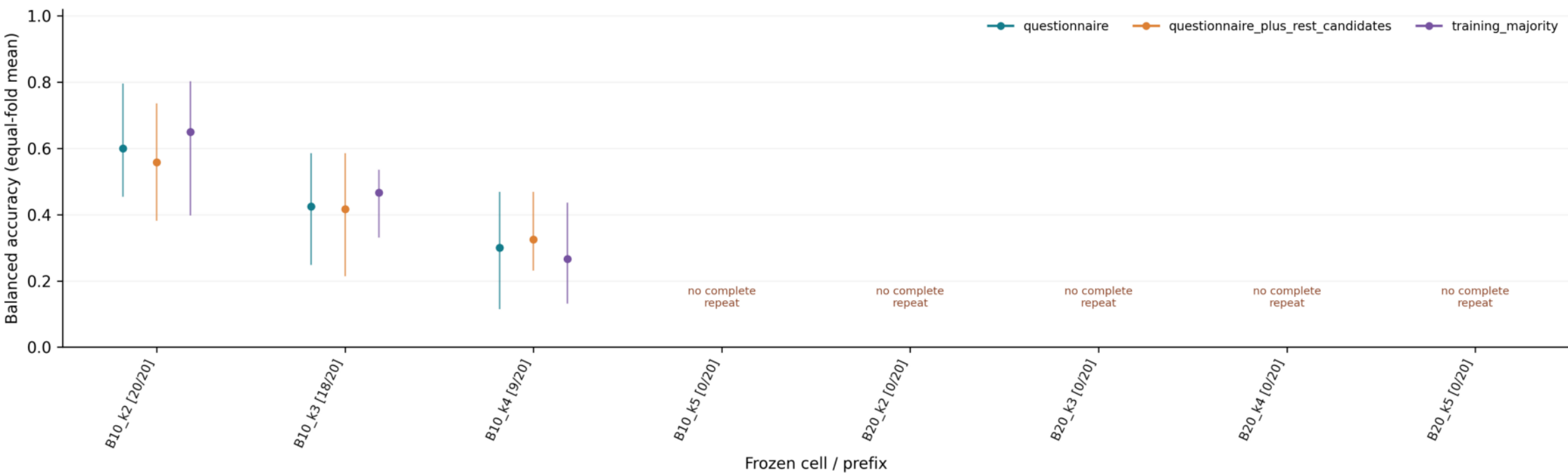
Algebraic bounds are not imputation or CIs. MC describes endpoint calculation precision, not scientific validity.

Supplement S5 | Training-partition recovery and baseline development

A Prefix and trained-shapelet recovery



B Baseline development: 1-10 min n=21; 1-20 min n=6; sealed closed



20 repeats attempted; ranges conditional on complete repeats. Undefined/incomplete retained; no clinical phenotypes or CIs.